Project Title: **Face Mask Detection Using Deep Learning**

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Introduction

Face mask detection is becoming more important, particularly in safety and health monitoring systems. In situations where safety regulations must be followed, the ability to automatically detect whether someone is wearing a mask is critical.

Traditional methods rely on manual monitoring or rule-based image processing, which are often inefficient. Recent advances in deep learning have enabled models to learn complex facial representations, leading to more accurate and reliable detection.

Problem statement

Face mask detection remains challenging due to variations in lighting, face orientation, occlusion, and mask appearance. Conventional methods struggle to effectively handle such variability.

This project uses deep learning techniques to automatically classify masked and unmasked faces. By leveraging transfer learning and convolutional neural networks, the proposed system seeks to develop a robust and accurate face mask detection model.

Deep learning is particularly suitable for this task due to its ability to extract high-level visual features while also handling complex real-world variations in images.

Objectives

1.Implement a deep learning model to detect whether a person is wearing a mask or not.

2.Improve the model by training it using images of people wearing masks and people not wearing masks.

3.Evaluate the model performance by analyzing the confusion matrix, training and validation accuracy/loss curves, and F1-score.

4.Analyze and interpret the results to assess the model's effectiveness in detecting face masks.

Dataset and Data Preparation

The selected dataset for this project is the Face Mask 12K Images Dataset obtained from Kaggle (Jangra, 2020). PipelineThis dataset contains approximately 12,000 labeled facial images categorized into two classes: individuals wearing face masks and individuals not wearing face masks. The dataset is organized into separate training, validation, and testing subsets with a balanced distribution between the two classes.

This dataset was chosen due to its adequate size, structured organization, and suitability for binary image classification tasks. The images represent real-world variations in lighting conditions, facial orientations, background complexity, and mask types, which enhances the model's ability to generalize effectively to unseen data.

Planned methodology/architecture

This project adopts a deep learning approach for face mask classification using Convolutional Neural Networks (CNNs) and transfer learning.

Previous studies implemented real-time object detection for mask detection. However, this project focuses on image classification, where images are analyzed to determine whether a person is wearing a mask or not.

A pre-trained MobileNetV2 model is used as the base model to extract high-level features from the images.

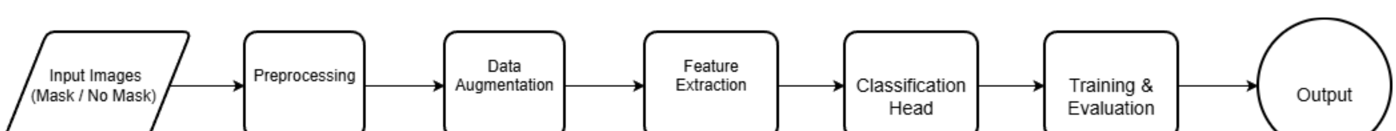
The system **pipeline** consists of the following steps:

Input & Preprocessing: The system receives face images labeled with mask or without mask. Images are resized, normalized, converted to arrays, and labels are one-hot encoded. Data augmentation (rotation, flipping, zooming, brightness) is applied to improve performance and reduce overfitting.

Feature Extraction & Classification: A pre-trained CNN model (MobileNetV2) extracts high-level features. A classification head with Pooling, Flatten, and Dense layers predicts mask or no mask.

Training & Evaluation: The model is fine-tuned using an optimization algorithm over multiple epochs. Performance is evaluated using standard evaluation metrics such as accuracy and F1-score, along with confusion matrix analysis and loss curves.

Output: The model outputs the expected category indicating whether the person is wearing a mask or not.



The proposed deep learning pipeline for face mask detection using MobileNetV2 and transfer learning

Expected Outcomes

The project is expected to use a pre-trained MobileNetV2 model as the basis for classifying face masks. Transfer learning will be applied by adjusting the final classification layer to categorize images into two groups: with and without a mask.

The model will be evaluated using the same evaluation criteria mentioned in the original study, including accuracy, F1- score, loss curves, and confusion matrix analysis.

Various optimization techniques will be applied and compared to the base model to analyze their impact on performance and determine the most effective approach.

Relevant references

1- Jangra, A. (2020). *Face Mask 12K Images Dataset*. Kaggle. https://www.kaggle.com/datasets/ashishjangra27/face-mask-12k-images-dataset

2 - Nagrath, P., Jain, R., Madan, A., Arora, R., Kataria, P., & Hemanth, D. J. (2021). SSDMNV2: A real-time CNN-based face mask detection system using MobileNetV2. *Sustainable Cities and Society, 66*, 102692. https://doi.org/10.1016/j.scs.2020.102692